

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

September 2025 – May 2027 (Expected)

Amherst, MA

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, Go, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend & APIs	FastAPI, Node.js, Express, REST APIs, microservices, async / concurrent services, multi-tenant systems, integration testing
Distributed & Scale	Distributed-data processing, Apache Spark / PySpark, high-throughput / low-latency tuning, message passing (Kafka, AWS SQS)
Cloud & DevOps	AWS (SQS, S3, KMS), Docker, Kubernetes, GCP, Azure, Git, GitHub Actions, CI/CD, Linux
Databases	PostgreSQL, DynamoDB, Redis, MongoDB, SQLite
ML & Computer Vision	PyTorch, CNNs / ResNet, image classification, transfer learning, LLM agents, OpenAI / Groq APIs

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions**, integrated with an external partner platform via REST APIs, aggregation pipelines, and **AWS SQS** asynchronous message passing; contributed to **\$340K+** in revenue.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Owned backend reliability across the eTriage stack: monitored production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

github.com/Deva-1903/ciir_agentic_search

- Developed multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and reviewer-facing trust user interface (UI) for inspecting extraction quality and ranking transparency.

Spark ETL - Distributed Data Processing at Scale

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

github.com/Deva-1903/Spark-ETL-Optimization

- Built a distributed data pipeline over a **1.4B-row** dataset, cutting end-to-end runtime **25%** via broadcast joins, adaptive query execution, and partition tuning for **high-throughput** processing.
- Reduced analytics **tail-latency** ratio (P99/P50) from **2.14x to 1.74x** via partition/column pruning and filter pushdown, profiling stage timings via Spark UI.

cf-ai-research-scout - Edge-Native AI Research Assistant

TypeScript, Cloudflare Workers AI, WebSockets, SQL

github.com/Deva-1903/cf_ai_research_scout

- Built edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses back to the client over **WebSocket** in real time, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.